



```
In [2]: import numpy as np
import pandas as pd
import seaborn as sns
```

```
In [4]: df = pd.read_csv("IRIS.csv")
print("First 5 rows of dataset:")
print(df.head())
```

First 5 rows of dataset:

|   | sepal_length | sepal_width | petal_length | petal_width | species     |
|---|--------------|-------------|--------------|-------------|-------------|
| 0 | 5.1          | 3.5         | 1.4          | 0.2         | Iris-setosa |
| 1 | 4.9          | 3.0         | 1.4          | 0.2         | Iris-setosa |
| 2 | 4.7          | 3.2         | 1.3          | 0.2         | Iris-setosa |
| 3 | 4.6          | 3.1         | 1.5          | 0.2         | Iris-setosa |
| 4 | 5.0          | 3.6         | 1.4          | 0.2         | Iris-setosa |

```
In [6]: print("\nStatistical Summary:")
print(df.describe())
```

Statistical Summary:

|       | sepal_length | sepal_width | petal_length | petal_width |
|-------|--------------|-------------|--------------|-------------|
| count | 150.000000   | 150.000000  | 150.000000   | 150.000000  |
| mean  | 5.843333     | 3.054000    | 3.758667     | 1.198667    |
| std   | 0.828066     | 0.433594    | 1.764420     | 0.763161    |
| min   | 4.300000     | 2.000000    | 1.000000     | 0.100000    |
| 25%   | 5.100000     | 2.800000    | 1.600000     | 0.300000    |
| 50%   | 5.800000     | 3.000000    | 4.350000     | 1.300000    |
| 75%   | 6.400000     | 3.300000    | 5.100000     | 1.800000    |
| max   | 7.900000     | 4.400000    | 6.900000     | 2.500000    |

```
In [23]: df.species.unique()
```

```
Out[23]: array(['Iris-setosa', 'Iris-versicolor', 'Iris-virginica'], dtype=object)
```

```
In [25]: df.groupby(df.species)
```

```
Out[25]: <pandas.core.groupby.generic.DataFrameGroupBy object at 0x00000209ACEA5880>
```

```
In [27]: df.groupby(df.species).count()
```

```
Out[27]:
```

|                        | sepal_length | sepal_width | petal_length | petal_width |
|------------------------|--------------|-------------|--------------|-------------|
| <b>species</b>         |              |             |              |             |
| <b>Iris-setosa</b>     | 50           | 50          | 50           | 50          |
| <b>Iris-versicolor</b> | 50           | 50          | 50           | 50          |
| <b>Iris-virginica</b>  | 50           | 50          | 50           | 50          |

```
In [29]: df.groupby(df.species).max()
```

```
Out[29]:
```

|                 | sepal_length | sepal_width | petal_length | petal_width |
|-----------------|--------------|-------------|--------------|-------------|
| species         |              |             |              |             |
| Iris-setosa     | 5.8          | 4.4         | 1.9          | 0.6         |
| Iris-versicolor | 7.0          | 3.4         | 5.1          | 1.8         |
| Iris-virginica  | 7.9          | 3.8         | 6.9          | 2.5         |

```
In [31]: df.groupby(df.species).min()
```

```
Out[31]:
```

|                 | sepal_length | sepal_width | petal_length | petal_width |
|-----------------|--------------|-------------|--------------|-------------|
| species         |              |             |              |             |
| Iris-setosa     | 4.3          | 2.3         | 1.0          | 0.1         |
| Iris-versicolor | 4.9          | 2.0         | 3.0          | 1.0         |
| Iris-virginica  | 4.9          | 2.2         | 4.5          | 1.4         |

```
In [33]: df.groupby(df.species).mean()
```

```
Out[33]:
```

|                 | sepal_length | sepal_width | petal_length | petal_width |
|-----------------|--------------|-------------|--------------|-------------|
| species         |              |             |              |             |
| Iris-setosa     | 5.006        | 3.418       | 1.464        | 0.244       |
| Iris-versicolor | 5.936        | 2.770       | 4.260        | 1.326       |
| Iris-virginica  | 6.588        | 2.974       | 5.552        | 2.026       |

```
In [35]: df.groupby(df.species).std()
```

```
Out[35]:
```

|                 | sepal_length | sepal_width | petal_length | petal_width |
|-----------------|--------------|-------------|--------------|-------------|
| species         |              |             |              |             |
| Iris-setosa     | 0.352490     | 0.381024    | 0.173511     | 0.107210    |
| Iris-versicolor | 0.516171     | 0.313798    | 0.469911     | 0.197753    |
| Iris-virginica  | 0.635880     | 0.322497    | 0.551895     | 0.274650    |

```
In [41]: df.groupby(df.species).describe()
```

Out[41]:

|                 |       |       |          |     |       | sepal_length |     |     | sepal_width |       | .. |
|-----------------|-------|-------|----------|-----|-------|--------------|-----|-----|-------------|-------|----|
|                 | count | mean  | std      | min | 25%   | 50%          | 75% | max | count       | mean  | .. |
| species         |       |       |          |     |       |              |     |     |             |       |    |
| Iris-setosa     | 50.0  | 5.006 | 0.352490 | 4.3 | 4.800 | 5.0          | 5.2 | 5.8 | 50.0        | 3.418 | .. |
| Iris-versicolor | 50.0  | 5.936 | 0.516171 | 4.9 | 5.600 | 5.9          | 6.3 | 7.0 | 50.0        | 2.770 | .. |
| Iris-virginica  | 50.0  | 6.588 | 0.635880 | 4.9 | 6.225 | 6.5          | 6.9 | 7.9 | 50.0        | 2.974 | .. |

3 rows × 32 columns

In [ ]: